

AI-First Fraud and Risk Management in U.S. Digital Banking

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Abstract— Digital banking fraud continues to increase as real-time payment systems accelerate transaction velocity and reduce review windows. Rule-based fraud systems detect known patterns but lack adaptability to evolving attacks such as synthetic identities, mule networks, and automated credential abuse. This short paper presents a compact deep-learning baseline model using a Multi-Layer Perceptron (MLP) trained on the IEEE-CIS fraud dataset. The model achieves strong ranking performance and demonstrates how threshold tuning, cost-sensitive evaluation, and calibration translate statistical outputs into operational and financial decisions. The study serves as an analytical foundation for AI-first fraud detection frameworks in U.S. banking.

I. INTRODUCTION

Fraud losses in U.S. digital banking exceed billions annually, driven by increasingly automated and sophisticated attacks. Real-time transactions limit the time available for rule-based verification, making machine learning approaches essential for detecting behavioral anomalies. This work evaluates whether a lightweight neural baseline can produce reliable fraud-risk signals and support decision-making through calibrated probabilities and expected-loss optimization.

II. METHODOLOGY

The IEEE-CIS dataset contains ~590K transactions with ~0.3% fraud prevalence. After cleaning, encoding, and feature preparation, 186 variables were used as model inputs. The final MLP architecture includes Dense layers with 50, 25, and 12 units, followed by dropout and batch normalization, yielding **10,998 trainable parameters**. The model was trained with the Adam optimizer and binary cross-entropy loss. No oversampling was applied to preserve a true operational baseline. Evaluation metrics include ROC-AUC, Precision-Recall (PR), confusion matrix, threshold sensitivity, expected loss, and calibration.

III. RESULTS

The model achieves a **ROC-AUC of 0.8649**, indicating strong ability to rank fraudulent transactions above legitimate ones. The PR curve further demonstrates reasonable precision at moderate recall levels in an imbalanced setting. At the default 0.50 threshold, precision is high but recall remains low, showing that accuracy alone is not meaningful for fraud detection. Threshold-sensitivity analysis shows that lowering the decision threshold substantially increases recall, reducing missed fraud. Expected-loss evaluation indicates that **threshold = 0.10 minimizes total financial loss**, reflecting the high cost of false negatives compared to false positives. The calibration curve shows predicted probabilities generally align with true fraud likelihood, supporting their use in risk-tiering.

IV. CONCLUSION

This study demonstrates that a compact MLP can provide meaningful fraud-risk signals even in an anonymized, highly imbalanced dataset. While insufficient for production deployment alone, the model forms a strong analytical baseline for AI-first fraud detection frameworks. Future work should incorporate cost-sensitive learning, graph-based fraud modeling, anomaly detection, and model-governance components to meet real-world regulatory and operational requirements.